

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

DIETING SYSTEM BASED ON DISEASE

Aim:

To implement a Prolog-based diet recommendation system according to a person's disease.

Objective:

To recommend suitable food items based on disease-related rules.

Algorithm:

- 1. Start.**
- 2. Define disease facts.**
- 3. Define recommended foods for each disease.**
- 4. Query the person's disease.**
- 5. Retrieve recommended foods.**
- 6. Display the diet recommendation.**
- 7. Stop.**

Flowchart

START

| v

Enter Disease

| v

Check Disease Database

| v

Find Recommended Food

| v

Display Diet

| v

STOP

Prolog Program:

% Diet Recommendation System

disease(diabetes). disease(hypertension). disease(obesity).

recommended_food(diabetes, vegetables).

recommended_food(diabetes, whole_grains).

recommended_food(diabetes, low_sugar_fruits).

recommended_food(hypertension, vegetables).

recommended_food(hypertension, low_sodium_food).

recommended_food(hypertension, fruits).

recommended_food(obesity, vegetables). recommended_food(obesity, salads).

recommended_food(obesity, whole_grains).

diet_for(Disease, Food) :-disease(Disease),

recommended_food(Disease, Food).

Queries

?- diet_for(diabetes, Food).

Output:

Result:

Thus, the Prolog program successfully recommends suitable food items based on the given disease.